



قواعد البيانات المتقدمة

12:00-1:00

22/11/2023

د/ ابراهيم العوضي

Faculty of Computers & Information, Assiut University

Course code: IS411

4th Level

Midterm Exam

Duration: One hour

This exam for the following program(s) : (Add the program's name(s))

* Required

* This form will record your name, please fill your name.

1

* الإسم الرباعي (بالعربي فقط)

2

رقم الجلوس *

3

المستوي *

- ☐ الاول
- ☐ الثاني
- ☐ الثالث
- ☐ الرابع 2013
- ☐ الرابع 2014
- ☐ الرابع 2015
- ☐ الرابع 2016
- ☐ الرابع 2017
- ☐ الرابع 2018
- ☐ الرابع 2019
- ☐ الرابع 2020

4

* البرنامج

- ☐ عام
- ☐ بايو
- ☐ هندسة
- ☐ ذكاء صناعي

5

* رقم المعمل

- ☐ ج٠
- ☐ د٠
- ☐ ه٠
- ☐ ب١
- ☐ ج١
- ☐ د١
- ☐ ه١
- ☐ أ٢
- ☐ ب٢
- ☐ ج٢
- ☐ د٢
- ☐ ه٢
- ☐ أ٣
- ☐ ب٣
- ☐ ج٣
- ☐ د٣
- ☐ ه٣
- ☐ أ٤

☐ ب ٤

☐ ج ٤

☐ د ٤

☐ هـ ٤

6

* رقم الكمبيوتر

7

* الكود (قد تمت مراجعة بيانات الطالب ورقم الجلوس)

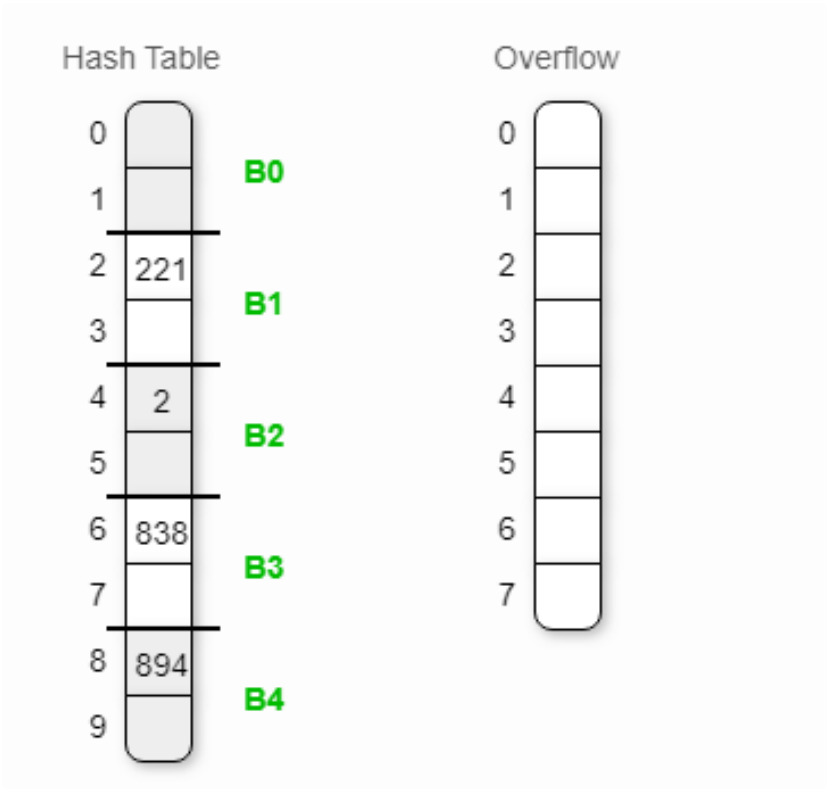
8

The main problem in the hash table is (1 Point)

- ☐ fixed size
- ☐ that many items can have the same key
- ☐ that even the serach time is $O(1)$, but in reality, the order is $O(n)$

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You are given a hash table of 5 buckets, each of size 2. Using the **first** bucket hash method described above, put key value 48 into the hash table. (1 Point)



- ☐ HashTable[0]
- ☐ HashTable[1]
- ☐ HashTable[2]

☐ HashTable[3]☐ HashTable[4]☐ HashTable[5]☐ HashTable[6]☐ HashTable[7]☐ HashTable[8]☐ HashTable[9]☐ Overflow[0]☐ Overflow[1]☐ Overflow[2]

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ASCII code

Dec	Hex	Name	Char	Ctrl-char	Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char
0	0	Null	NUL	CTRL-@	32	20	Space	64	40	@	96	60	`
1	1	Start of heading	SOH	CTRL-A	33	21	!	65	41	A	97	61	a
2	2	Start of text	STX	CTRL-B	34	22	"	66	42	B	98	62	b
3	3	End of text	ETX	CTRL-C	35	23	#	67	43	C	99	63	c
4	4	End of xmit	EOT	CTRL-D	36	24	\$	68	44	D	100	64	d
5	5	Enquiry	ENQ	CTRL-E	37	25	%	69	45	E	101	65	e
6	6	Acknowledge	ACK	CTRL-F	38	26	&	70	46	F	102	66	f
7	7	Bell	BEL	CTRL-G	39	27	'	71	47	G	103	67	g
8	8	Backspace	BS	CTRL-H	40	28	(	72	48	H	104	68	h
9	9	Horizontal tab	HT	CTRL-I	41	29	)	73	49	I	105	69	i
10	0A	Line feed	LF	CTRL-J	42	2A	*	74	4A	J	106	6A	j
11	0B	Vertical tab	VT	CTRL-K	43	2B	+	75	4B	K	107	6B	k
12	0C	Form feed	FF	CTRL-L	44	2C	,	76	4C	L	108	6C	l
13	0D	Carriage feed	CR	CTRL-M	45	2D	-	77	4D	M	109	6D	m
14	0E	Shift out	SO	CTRL-N	46	2E	.	78	4E	N	110	6E	n
15	0F	Shift in	SI	CTRL-O	47	2F	/	79	4F	O	111	6F	o
16	10	Data line escape	DLE	CTRL-P	48	30	0	80	50	P	112	70	p
17	11	Device control 1	DC1	CTRL-Q	49	31	1	81	51	Q	113	71	q
18	12	Device control 2	DC2	CTRL-R	50	32	2	82	52	R	114	72	r
19	13	Device control 3	DC3	CTRL-S	51	33	3	83	53	S	115	73	s
20	14	Device control 4	DC4	CTRL-T	52	34	4	84	54	T	116	74	t
21	15	Neg acknowledge	NAK	CTRL-U	53	35	5	85	55	U	117	75	u
22	16	Synchronous idle	SYN	CTRL-V	54	36	6	86	56	V	118	76	v
23	17	End of xmit block	ETB	CTRL-W	55	37	7	87	57	W	119	77	w
24	18	Cancel	CAN	CTRL-X	56	38	8	88	58	X	120	78	x
25	19	End of medium	EM	CTRL-Y	57	39	9	89	59	Y	121	79	y
26	1A	Substitute	SUB	CTRL-Z	58	3A	:	90	5A	Z	122	7A	z
27	1B	Escape	ESC	CTRL-[	59	3B	;	91	5B	[	123	7B	{
28	1C	File separator	FS	CTRL-\	60	3C	<	92	5C	\	124	7C	
29	1D	Group separator	GS	CTRL-]	61	3D	=	93	5D	]	125	7D	}
30	1E	Record separator	RS	CTRL-^	62	3E	>	94	5E	^	126	7E	~
31	1F	Unit separator	US	CTRL-~	63	3F	?	95	5F	_	127	7F	DEL

11

Database file is considered as file of record, and we read these records in one step and store them again in another step (1 Point)

☐ True☐ False

12

The data structures used in database are data structure (1 Point)

☐ File☐ RAM☐ Both RAM and file

13

In Physical Design, we consider the database is a set of (1 Point)

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SELECT *
FROM EMP;

The best data storage for EMP is (1 Point)

- ☐ N-ary Storage Model
- ☐ Decomposition Storage Model
- ☐ Hybrid Storage Model

15

SELECT Average(degree)
FROM Exam;

The best data storage for Exam is (1 Point)

- ☐ N-ary Storage Model
- ☐ Decomposition Storage Model
- ☐ Hybrid Storage Model

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The data storage models used in DBMS are (1 Point)

- ☐ N-ary Storage Model
- ☐ Decomposition Storage Model
- ☐ Hybrid Storage Model
- ☐ Others

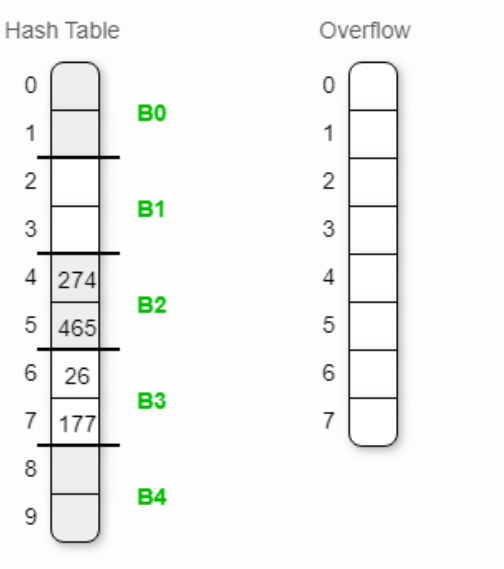
17

The last character in a linux file is (1 Point)

- ☐ \0
- ☐ EOF
- ☐ ctrl+d
- ☐ ctrl+z

18

You are given a hash table of 5 buckets, each of size 2. Using the **first** bucket hash method described above, put key value 3 into the hash table. (1 Point)



- ☐ HashTable[0]
- ☐ HashTable[1]
- ☐ HashTable[2]
- ☐ HashTable[3]
- ☐ HashTable[4]
- ☐ HashTable[5]
- ☐ HashTable[6]
- ☐ HashTable[7]
- ☐ HashTable[8]
- ☐ HashTable[9]
- ☐ Overflow[0]
- ☐ Overflow[1]

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Assume that we use dynamic hash table with two bits, the word Egypt is stored in hash table number (1 Point)

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The hash table used in some DBMS is called (1 Point)

- ☐ Dynamic hash table
- ☐ Bucket hash table
- ☐ fixed size hash table with overflow
- ☐ Linked list to overcome collision

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The item in hash table is (1 Point)

22

What is the hash value for Egypt, with a simple hash function? (1 Point)

23

To print an interger variable, which hase the name number, in a binary file, we use (1 Point)

- ☐ fprintf(fPtr, "%d", number);
- ☐ fwrite(&number, sizeof(int), 1, fPtr);
- ☐ fwrite(fPtr, "%d", number);

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